

Rental Housing Law Navigator

RealPage x Hack-Nation, Challenge 2 | Live: navigator.mdsrana.com | Code: github.com/Sheru01/rental-law-navigator

"It tells you what it cannot tell you, and exactly why."

1 Challenge Tackled

Which housing rules apply at a given address, on a given date? Housing law is layered (state over city), each layer with its own coverage tests and effective dates, and the public record often lacks the one building fact a rule turns on. Users: housing agencies, tenant advocates, housing providers, and anyone deploying AI on law. We built an engine that answers by address and date over a 500-address, three-state sample, and that treats an honest "unknown" as a first-class result rather than a failure.

2 Tools / ML Models Used

- **LLM extraction pipeline** (Python, swappable OpenAI / Anthropic / replay transports): turns corpus legal text into structured rule records with per-field epistemics. Demo data uses replayed extraction fixtures, not a live model run, and the site header says so.
- **Claude agents** (Opus, Sonnet, Fable via Claude Code) built, audited and reviewed pipeline, engine, site and submission; a second blind reader (GPT) cross-checked extractions and disagreements were published, not reconciled.
- **US Census Geocoder** (live, two passes: address batch, then per-row coordinates lookup) resolves each address to its incorporated place, because the batch endpoint does not return it and the mailing city is not the governing jurisdiction.
- **Deterministic three-valued engine** (Python stdlib): coverage, precedence (yields_to / stacks / preempts_pending), as-of dates, withheld-status projection. No LLM decides applicability.
- **Static site generator + Playwright**: one self-contained page from engine output; an 11-check browser smoke suite gates every push; deployed on Vercel.

3 What Worked Well

- Live geocoding of all 500 addresses: 479 resolved to an incorporated place, 21 reported unresolved rather than guessed; the project validator passes with 0 warnings.
- Span discipline: every shipped rule carries a quoted span that exact-matches its source document after whitespace normalization, re-verified in the browser at render time (7 of 7 pass).
- Two kinds of silence kept separate: **unknown** (a building fact is missing) vs **indeterminate** (the legal record cannot establish the rule is operative). Aggregated, they form the Evidence Debt Ledger: a sourced map of what the public record cannot answer.
- Change tracking as a parameter, not a feature: moving the as-of date flips NJ's FAIR Act from not-yet-effective to applies on its derived 2027-07-01 date, shown as a diff.

4 What Was Challenging

- Geocoder and engine disagreed on a key name (legal_city vs jurisdiction), so every address read as unresolved the first time real data flowed. Caught by binding real output instead of fixtures; fixed with the full 273-test suite green.
- The Census batch endpoint omits the incorporated place entirely; we added a second live call per matched row.
- Documents we could not source as legal text (fetch-tool summaries) are hatched, down-weighted, and never quoted as law; two rules (Berkeley ordinance, CA AB 325) ship as withheld because their operative status is not establishable from the corpus.

5 How We Spent the Time (solo + AI agents)

H0-4: corpus survey, frozen contract (ids, coverage schema, precedence, span rules). H4-10: parallel agent builds (extraction pipeline, geocode module, engine) plus an eight-board demo design under a design freeze: no verdict hard-coded for the mockup. H10-14: live Census run, read-only audit, engine fix, 273 tests green. H14-18: site generated from validated output only, deployed to navigator.mdsrana.com. H18-20: visual redesign, smoke suite, submission package.

If we had 24 more hours: run the live model extraction across all 74 documents and publish the two-reader disagreement table as a first-class page.